

Industrialization

Between about 1750 and 1850, Britain transformed itself from an agricultural nation into the world's top industrial power. Cloth making was its most profitable enterprise. At this time, many workers moved from farms into the new factories, where new machines were powered by steam—a period known as the Industrial Revolution.

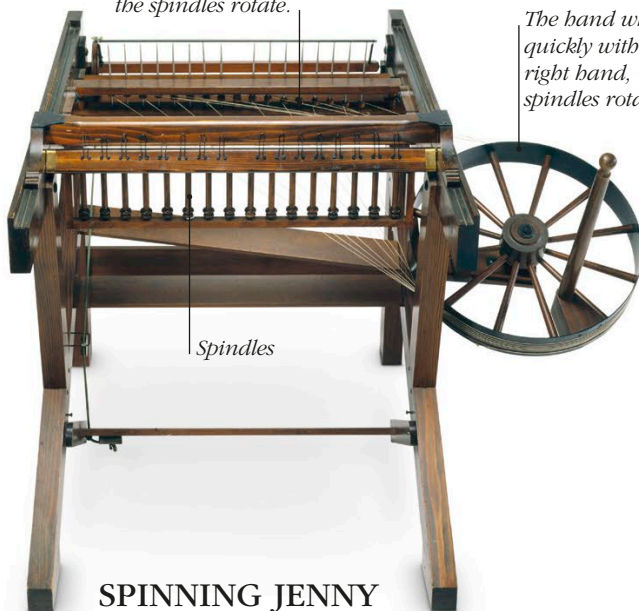
WOW!

Not everyone welcomed the Industrial Revolution: Luddites were workers who destroyed machinery to protest change.

THE STEAM ENGINE

Steam was the world's first great power source. In 1711, English engineer Thomas Newcomen built the first steam engine (see p.56), which was based on an earlier steam pump by Thomas Savery. Newcomen's machine, however, was inefficient, and it was not until Scotsman James Watt made improvements and patented his own engine in 1769 (right) that steam engines were made to power machinery.

Thread is spun with the spinner's left hand as the spindles rotate.



Spindles

The hand wheel is turned quickly with the spinner's right hand, making the spindles rotate.

SPINNING JENNY

Traditionally, workers operated spinning wheels, which could spin only one thread at a time. James Hargreaves's Spinning Jenny of 1764 could spin several threads at once. Along with the steam engine, it helped to kickstart the Industrial Revolution in Britain.

Piston rod

Steam in the cylinder pushes up the piston connected to the end of the beam by the piston rod.

